COMP3013 DBMS Lab 6

United International College

Data Definition

- In lectures, we have introduced conceptual designs (ER diagrams) and logical designs (schemas).
- In this lab, we will implement the logical design in MySQL.
- **Data definition language** defines
  - Databases
  - Tables
  - Attributes and their types
  - Constraints

Create Database

```
CREATE DATABASE uic_example
```

- Suppose the database is called “uic_example”.
- Execute the query. Then, the new database will appear in the database list.

Create Tables

- Before creating tables, let's look at the schemas from Lecture 6.

$programs = (\underline{p\_code}, p\_name, division, director)$

$course = (\underline{c\_name}, credits, domain, c\_number)$

$offer = (\underline{p\_code, c\_name})$

$section = (\underline{c\_name, s\_number, sem}, venue, time, instructor\_id)$

$enroll = (\underline{id, c\_name, s\_number, sem}, grade)$

$students = (\underline{id}, s\_name, year, gpa, major)$

$instructors = (\underline{id}, i\_name, title, salary, program)$

$contact = (\underline{id, phone})$

$books = (\underline{ISBN}, b\_title, author)$

$borrow = (\underline{id, ISBN}, return, borrow)$

Create Tables

- To create tables, we use

```
CREATE TABLE table_name (
    attribute1 type1,
    attribute2 type2,
    …
    constraint1,
    constraint2,
    …
)
```

- "`table_name`" is the name of the table.
- "`attribute`" is the name of the attribute.
- "type" is the data type for the attribute.
- "constraint" is a condition on the table. If users try to insert some data violating the condition, the system will give a warning.

Example

Table name

Attribute name

```
CREATE TABLE book(
    ISBN    INT,
    b_title VARCHAR(40)  NOT NULL,
    author  VARCHAR(20),
    PRIMARY KEY (ISBN)
)
```

Data types   Constraints

- Remember to select the database “UIC” before executing the query.

Data Types

- **`char(n)`** fixed length character string of at most length n.
- **`varchar(n)`** variable length character string of at most length n.
- **`int`** integer (a finite subset of the integers that is machine-dependent).
- **`smallint`** small integer (a machine-dependent subset of the integer domain type).
- **`numeric(p,n)`** fixed point number, with user-specified precision of p digits, and with n digits to the right of decimal point.
- **`real`** floating point, with machine-dependent precision.
- **`float(n)`** floating point number, with user-specified precision of at least n digits.
- **`year`** 4-digit string or 4-digit number from 1901 to 2155.
- **`date`** in 'YYYY-MM-DD' format from 1000-01-01 to 9999-12-31.
- **`time`** in 'HH:MM:SS' format.
- **`blob`** binary large object, usually for images.
- **`clob`** character large object, usually for long text.

Constraints

- **`PRIMARY KEY (attribute)`**
  - One or multiple attribute(s)
  - Uniquely identify the tuples
  - Unique for each table
  - Cannot be a NULL value by default
- **`NOT NULL`**
  - The value of the attribute cannot be **unknown**.
  - The condition is checked when rows are inserted into the table.

Example

- Define the table for “borrow”.

```
CREATE TABLE borrow(
    id              INT  NOT NULL,
    ISBN            INT  NOT NULL,
    return_date     DATE,
    borrow_date     DATE NOT NULL,
    PRIMARY KEY (id, ISBN)
)
```

Foreign key

$book = (\underline{ISBN}, b\_title, author)$

$borrow = (\underline{id, ISBN}, return\_date, borrow\_date)$

- A value of "ISBN" in "borrow" cannot exist if such an ISBN is not in "book". (How do you borrow a book which does not exist?)
- Thus, "ISBN" in "borrow" is a foreign key referencing to "ISBN" in "book".
- When users try to borrow a book (insert a row to table "borrow"), the system should check the existence.

- **`FOREIGN KEY (attribute(s)1) REFERENCES table2(attribute(s)2)`**
  - "attribute(s)1" in the current table and "attribute(s)2" in "table2" represent the same piece of information.
  - "attribute(s)2" is the primary key for "table2".
  - A value of "attribute(s)1" in the current table **cannot exist alone**.
  - More will be on this in the following labs.

Foreign key

- To add foreign keys to an existing table, we need to change the schema.

```
ALTER TABLE borrow ADD FOREIGN KEY (ISBN) REFERENCES books(ISBN)
```

- `ALTER TABLE`: the keyword to change the schema of a table.
- `borrow`: the table name.
- `ADD`: to add new attributes or constraints (“`DROP`” to remove attributes or constraints).
- `FOREIGN KEY`: the constraint.
- `(ISBN)`: the foreign key.
- `REFERENCES`: the keyword to indicate which attribute is the target.
- `books(ISBN)`: the attribute “ISBN” in the table “books” is referenced.

Exercises

Implement the logical design in MySQL

- Create one table for each of the schema.
- Define the primary key.
- Define the suitable data type for each attribute.
- Define the foreign keys.

Save your queries in a txt file. Rename it as "COMP3013 Lab6 ###.txt", where "###" is your student ID. And submit it on iSpace. The DDL is 24 hours after the lab.